

PROJECT REPORT

Script-Controlled ACL – Restrict Record Access Based on Field Value

1. INTRODUCTION

1.1 Project Overview

The **Script-Controlled ACL – Restrict Record Access Based on Field Value** project demonstrates how Access Control Lists (ACLs) can be used in ServiceNow to provide record-level security.

The project creates an **Institution Details** table containing student information such as Student Roll Number, Student Name, Faculty Name, Branch, Email, Phone Number and Description. Records are created with different branch values such as ECE, EEE and CSE.

The system uses user roles and ACL rules to control access to these records. EEE branch users are allowed to view EEE branch records, while administrators retain full access. Separate ACLs are implemented for Read, Create, Write and Delete operations.

1.2 Purpose

The purpose of this project is to implement a script-controlled ACL mechanism that restricts unauthorized access to Institution Details records.

The project evaluates the user's role and record field values to determine whether access should be allowed or denied. It demonstrates how READ, CREATE, WRITE and DELETE operations can be controlled using separate ACLs.

2. IDEATION PHASE

2.1 Problem Statement

ServiceNow records may contain information that should not be accessible to every user. Without proper record-level access control, unauthorized users may view, create, modify or delete records.

The project addresses the need for controlled access by restricting Institution Details records according to user roles and the Branch field. EEE users should be able to access EEE branch records, while administrators should retain full access.

Customer Problem Statement

- ✓ **I am:** A ServiceNow user accessing Institution Details records.
- ✓ **I'm trying to:** Access only the records I am authorized to view.
- ✓ **But:** Some records should be restricted from unauthorized users.
- ✓ **Because:** Access needs to be controlled based on user roles and the Branch field.
- ✓ **Which makes me feel:** Concerned about unauthorized access to records.

2.2 Empathy Map Canvas

SAYS

- “I should only access the records I am authorized to view.”
- “EEE branch records should be restricted from unauthorized users.”
- “I need secure access to Institution Details records.”

THINKS

- “My records should not be visible to unauthorized users.”
- “Access should depend on my assigned role and the Branch field.”
- “Administrators should have complete access.”

DOES

- Logs in to the ServiceNow instance.
- Opens Institution Details records.
- Attempts to view or perform permitted record operations.

FEELS

- Concerned about unauthorized record access.
- Secure when access is properly controlled.

- Confident when only authorized records are accessible.

PERSONA

Name: ServiceNow User

Role / User Type: Authorized User

2.3 Brainstorming

The project ideas were focused on improving record-level security in ServiceNow.

S.No	Team Member	Idea / Suggestion	Category	Group No.
1	Team Member 1	Restrict record access based on user role	Access Control	1
2	Team Member 2	Allow only EEE users to view EEE records	Record Security	1
3	Team Member 3	Use ACL script to control record access and Create, Read, Write and Delete operations	Script Control and CRUD Security	1
4	Team Member 4	Provide administrators with full access Use Branch field to restrict record visibility	Admin and Field based Security	1

Idea Prioritization

Rate each grouped idea on feasibility and importance, then select the final idea(s) to move forward with.

Group No.	Final Idea	Feasibility (High/Medium/Low)	Importance (High/Medium/Low)	Priority Selected (Yes/No)
1	Implement Script-Controlled ACL for role and branch-based record access	High	High	Yes
1	Control READ, CREATE, WRITE and DELETE operations using separate ACLs	High	High	Yes
1	Restrict EEE branch records and provide admin full access	High	High	Yes

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Phase of Journey

Stage 1 – Login & Access

Stage 2 – Open Institution Details & ACL Check

Stage 3 – View / Perform Authorized Operation

Actions

Stage 1

- Logs in to ServiceNow.
- Accesses the application.

Stage 2

- Opens Institution Details.
- Selects or searches for records.
- Attempts to access a record.

Stage 3

- Views permitted records.
- Creates, edits or deletes records according to assigned roles.

Touchpoint

- ServiceNow Login
- Institution Details table
- Record list and form
- ACL security check
- New, Edit and Delete options

Customer Thought

- “I need to access the Institution Details records.”
- “Am I authorized to access this record?”
- “My operations should depend on my assigned role.”

Customer Feeling

- Expecting secure access.
- Concerned about unauthorized access.
- Confident when access is properly controlled.

Process Ownership

- ServiceNow
- System Administrator
- ACL configuration

Opportunities

- Restrict records based on role and Branch.
- Allow administrators full access.
- Apply separate ACLs for Read, Create, Write and Delete.

3.2 Solution Requirement

The system requires a custom **Institution Details** table named `u_institution_details`. The table contains Student Roll Number, Student Name, Faculty Name, Branch, Email, Phone Number and Description fields.

The system also requires the following custom roles:

- ✓ **bb1** – Read access
- ✓ **bb2** – Create access
- ✓ **bb3** – Write access
- ✓ **bb4** – Delete access

All four roles are assigned to the EEE User for testing.

The ACL requirements are:

- ✓ **Read ACL:** Branch must be EEE and the user requires bb1; administrators retain access.
- ✓ **Create ACL:** User requires bb2.
- ✓ **Write ACL:** User requires bb3.
- ✓ **Delete ACL:** User requires bb4.

3.3 Data Flow Diagram

The Data Flow Diagram represents the movement of data between the ServiceNow user, authentication process, role and Branch validation, ACL evaluation, and the Institution Details data store.

Main Flow

ServiceNow User → Authenticate User → Validate Role & Branch → Evaluate ACL Rules → Institution Details → Control Record Access → ServiceNow User

Data Flow

- Login Credentials

- User Role + Branch
- ACL Request
- Record Data
- Access Result – Allow / Deny

3.4 Technology Stack

The project specifically uses the ServiceNow ACL mechanism, custom roles and the u_institution_details table.

Technology / Component	Purpose
ServiceNow	Platform used to implement and test the ACL system
Access Control List (ACL)	Controls record-level access
ServiceNow Custom Table	Stores Institution Details records
ServiceNow Roles	Controls user permissions
ACL Script	Evaluates user role and access conditions
Institution Details Table	Stores student and branch-related records

4. PROJECT DESIGN

4.1 Problem Solution Fit

The project directly addresses the problem of unauthorized record access by combining **user roles, Branch values and ACL rules**.

The Read ACL allows EEE branch users to view EEE records, while users without the required role are denied access. Administrators can view all records regardless of Branch.

4.2 Proposed Solution

The proposed solution is to implement separate ACLs for the four major record operations:

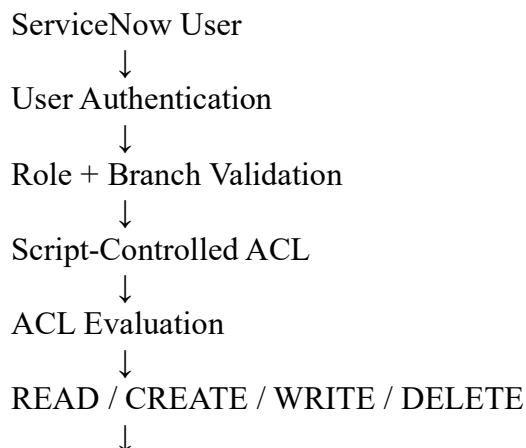
READ → CREATE → WRITE → DELETE

The Read ACL uses a script to check whether the user is an administrator or has the required role. The Create, Write and Delete operations are controlled through their respective ACL roles.

This provides controlled record-level access and prevents unauthorized users from performing restricted operations.

4.3 Solution Architecture

The solution architecture illustrates how the ServiceNow user interacts with the Institution Details table through role-based and script-controlled ACLs. The system validates the user's role and Branch value before allowing or denying access to records. Separate ACLs control Read, Create, Write and Delete operations, while administrators retain full access.



Institution Details Table



Access Decision



Allow / Deny

5. PROJECT PLANNING & SCHEDULING

Project Planning

Plan and organize the Script-Controlled ACL project by dividing the work into manageable tasks and sprints.

Sprint	Task	Story Points
Sprint-1	User, Role, Table and Record Creation	5
Sprint-2	READ and CREATE ACL Configuration	5
Sprint-3	WRITE and DELETE ACL Configuration	4
Sprint-4	Verification and Testing	3

Total Story Points : 17

Number of Sprints : 4

Duration : 9 Days

Project Outcome

The project implements script-controlled ACLs to control **READ, CREATE, WRITE, and DELETE** access to Institution Details records based on user roles and Branch values

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Functional Testing

Test Case	Test Scenario	Expected Result
FT-01	User with bb1 accesses Institution Details	Only EEE branch records are visible
FT-02	User without required role accesses records	Records are not accessible
FT-03	Administrator accesses Institution Details	All records are visible
FT-04	User with bb1 and bb2 checks Create access	New button is available
FT-05	User with bb1, bb2 and bb3 edits an EEE record	EEE record can be edited
FT-06	User with bb1, bb2, bb3 and bb4 deletes an EEE record	EEE record can be deleted

These expected behaviors are based on the project verification steps

Performance Testing

Test Case	Performance Scenario	Expected Result
PT-01	Measure ACL evaluation while opening records	Access evaluation should complete without noticeable delay
PT-02	Test access using multiple users	ACL rules should continue to be applied correctly
PT-03	Access multiple Institution Details records	Records should load successfully while maintaining access restrictions

Actual response times and Pass/Fail results should be entered after executing the tests.

7. RESULTS

7.1 Output Screenshots of Milestone 1

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AllFavoritesHistoryWorkspaces

User - New Record

Search

Submit

User

Save

Configure

Export

Create Favorite

Copy URL

Copy sys_id

Reload form

To save this record, save the record and then click Set Password.

User ID

EEE User

Email

eeeuser@gmail.com

Name

EEE

Language

-- None --

Last name

User

Calendar integration

Outlook

Title

Time zone

System (America/Los Angeles)

Department

Date format

System (yyyy-MM-dd)

Password needs reset

Business phone

Locked out

Mobile phone

Active

☑

Photo

Click to add...

Identity type

Human

Internal Integration User

Submit

servicenow

AllFavoritesHistoryWorkspaces

Users

Search

Actions on selected rows...

New

Users

Name

Search

All > Name >= eee user

User ID	Name	Email	Active	Created	Updated
EEE User	EEE User	eeeuser@gmail.com	true	2026-09-24 02:02:28	2026-09-24 02:02:28
efren.baucher	Efren Baucher	efren.baucher@example.com	true	2012-02-17 19:04:53	2026-09-15 06:29:30
elaine.renzi	Elaine Renzi	elaine.renzi@example.com	true	2012-02-17 19:04:52	2026-09-15 06:29:28
eldon.sutch	Eldon Sutch	eldon.sutch@example.com	true	2012-02-17 19:04:53	2026-09-15 06:29:37
eli.bettner	Eli Bettner	eli.bettner@example.com	true	2012-02-17 19:04:51	2026-09-15 06:29:31
elisa.gracely	Elisa Gracely	elisa.gracely@example.com	true	2012-02-17 19:04:52	2026-09-15 06:29:28
eliseo.wice	Eliseo Wice	eliseo.wice@example.com	true	2012-02-17 19:04:53	2026-09-15 06:29:29
ella.pahnke	Ella Pahnke	ella.pahnke@example.com	true	2012-02-17 19:04:49	2026-09-15 06:29:34
elmo.dagenais	Elmo Dagenais	elmo.dagenais@example.com	true	2012-02-17 19:04:53	2026-09-15 06:29:35
elmo.gabouer	Elmo Gabouer	elmo.gabouer@example.com	true	2012-02-17 19:04:53	2026-09-15 06:29:27
elvira.blumenthal	Elvira Blumenthal	elvira.blumenthal@example.com	true	2012-02-17 19:04:52	2026-09-15 06:29:33
emery.reek	Emery Reek	emery.reek@example.com	true	2012-02-17 19:04:49	2026-09-15 06:29:29
emilia.oxley	Emilia Oxley	emilia.oxley@example.com	true	2012-02-17 19:04:49	2026-09-15 06:29:34
emilio.lampkin	Emilio Lampkin	emilio.lampkin@example.com	true	2012-02-17 19:04:49	2026-09-15 06:29:37
enrique.oroark	Enrique Oroark	enrique.oroark@example.com	true	2012-02-17 19:04:50	2026-09-15 06:29:29

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AllFavoritesHistoryWorkspaces

Role - New Record

Search

Role

New record

Submit

* Name

bb1

Application

Global

Elevated privilege

☐

Description

Brief role purpose

Submit

servicenow

AllFavoritesHistoryWorkspaces

Role - New Record

Search

Role

New record

Submit

* Name

bb2

Application

Global

Elevated privilege

☐

Description

Brief role purpose

Submit

servicenow

AllFavoritesHistoryWorkspaces

Role - New Record

Search

RoleNew record

Submit

* Name

bb3

Application

Global

Elevated privilege

Description

Brief role purpose

Submit

servicenow

AllFavoritesHistoryWorkspaces

Role - New Record

Search

RoleNew record

Submit

* Name

bb4

Application

Global

Elevated privilege

Description

Brief role purpose

Submit

servicenow

AllFavoritesHistoryAdmin

User Role - Edit Members

Search

CancelSave

-- choose field --

-- oper --

-- value --

Collection

bb4

--None--

Roles List

EEE User

bb1bb2bb3bb4

CancelSave

Name

bb4

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AllFavoritesHistoryWorkspaces

User - EEE User

Search

UpdateSet PasswordDelete

User

EEE User

UpdateSet PasswordDelete

Related Links

[View linked accounts](#)[View Subscriptions](#)[Reset a password](#)

Entitled Custom Tables

Roles (5)Groups (1)DelegatesSubscriptionsUser Client Certificates

Role

Search

Actions on selected rows...Edit...

User = EEE User

Role	State	Inherited	Inheritance Count
bb1	Active	false	
bb2	Active	false	
bb3	Active	false	
bb4	Active	false	
snc_required_script_writer_permission	Active	true	1

1 to 5 of 5

Output Screenshots of Milestone 2

Table
New record

SubmitCancel

ServiceNow recommends creating custom tables in scoped applications. To learn more about creating scoped applications, click [here](#).

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)

* LabelInstitution details

* Nameu_institution_details

Extends table

ApplicationGlobal

Create module☒

Create mobile module☒

Add module to menu-- Create new --

New menu nameInstitution details

Remote Table☐

ColumnsControlsApplication Access

Table Columnsfor textSearch

Dictionary Entries

servicenowAll Favorites History AdminTable - Institution DetailsSearchDeleteUpdateDelete All Records

Column label	Type	Reference	Max length	Default value	Display
Sys ID	Sys ID (GUID)	(empty)	32		false
Created by	String	(empty)	40		false
Created	Date/Time	(empty)	40		false
Updated by	String	(empty)	40		false
Updates	Integer	(empty)	40		false
Updated	Date/Time	(empty)	40		false
Student Roll Number					false
Student Name		User			false
Faculty Name		User			false
Branch					false
Email	String				false
Phone Number	String				false
Description	String				false
Insert a new row...					

DeleteUpdateDelete All Records

Output Screenshots of Milestone 3

servicenow

Access Control - New Record

Search

Submit

Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.

* Type

record

* Operation

read

Decision Type

Allow If

Admin overrides

Protection policy

-- None --

Tables

* Name

Institution Details [u_institution_details]

-- None --

Description

Applies To

No. of records matching the condition: 4

Add Filter Condition

Add OR Clause

Application

Global

Active

Advanced

servicenow

AllFavoritesHistoryAdmin

Access Control - New Record

Search

Submit

Access Control

New record

Condition

All of these conditions must be met

-- choose field --

OR

AND

or

New Criteria

Data Condition

Condition

No. of records matching the condition: 4

Add Filter Condition

Add OR Clause

Branch

is

EEE

AND

OR

X

Advanced Condition

Controlled by References

Script

Turn on ECMAScript 2021 (ES12) mode

1

servicenow

AllFavoritesHistoryAdmin

Access Control - New Record

Search

Submit

Access Control
New record

Advanced Condition

Controlled by References

Script ☒ Turn on ECMAScript 2021 (ES12) mode

```
1 (function () {  
6   // Allow only EEE branch users to see EEE records  
7   if (gs.hasRole('bb1')){  
8     return true;  
9   }  
10  // Deny access for all others  
11  return false;  
12})();
```

Submit

servicenow

AllFavoritesHistoryWorkspaces

Access Controls

Search

System Administrator
System Administrator
security_admin
ServiceNow

Profile
Preferences
Keyboard shortcuts
Impersonate user
Elevate role
Printer friendly version
Log out

Access Controls

Name

Search

All

Name	Decision Type	Operation	Target	Active	Updated by	Updated
\$allappsmgmt					admin	2018-03-06 10:20:09
\$att_page_inspector					admin	2020-06-15 10:43:57
\$conversation-builder					admin	2016-01-19 12:53:23
\$mycompanyappsmgmt					admin	2024-02-18 06:04:39
\$oc_calendar_launcher_redirect					admin	2016-01-09 16:21:38
\$products					admin	
\$pwd_reset_serviceDesk					admin	
\$sa_service_model_json					admin	
\$sla_timeline					admin	
\$sn-vb-web-client-app					admin	
\$sn_global_search_results					admin	
\$spd	Allow If	read	ui_page	true	admin	
\$spm_availability_calculation_setting	Allow If	read	ui_page	true	admin	
\$studio	Allow If	read	ui_page	true	admin	

1 to 20 of 43,717

Impersonate user

Select a user

Search for a user

RECENT IMPERSONATIONS

EU EEE User
EEE User

Cancel Impersonate user

servicenow All Favorites History : Institution Details ☆						
Institution Details Branch Search Actions on selected rows...						
All						
☐	Branch	Description	Email	Faculty Name	Phone Number	Student Name
	EEE	EEE	johnsmith@gmail.com	Abraham Lincon	7869589087	John Smith
						Student Roll Number
						INR0001001
1 to 1 of 1						

Output Screenshots of Milestone 4

servicenow		All	Favorites	History	Admin	Access Control - New Record ☆		Search								
Access Control		New record														Submit
Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.																
* Type	record		Application		Global											
* Operation	read		Active		☒											
Decision Type	Allow If		Advanced		☐											
Admin overrides			☒													
Protection policy			-- None --													
* Name	Institution Details [u_institution_details]		-- None --													
Description																
Applies To			No. of records matching the condition: 4													
			Add Filter Condition Add OR Clause													
			-- choose field -- -- oper -- -- value --													
Conditions																

servicenow All Favorites History Admin : Access Control - New Record Search

Access Control New record Submit

Condition All of these conditions must be met

-- choose field --

OR AND

or

New Criteria

Data Condition

Condition No. of records matching the condition: 4

Add Filter Condition Add OR Clause

Branch is EEE AND OR X

Advanced Condition

Controlled by References

Script Turn on ECMAScript 2021 (ES12) mode

1

servicenow All Favorites History Admin : Access Control - New Record Search

Access Control New record Submit

Advanced Condition

Controlled by References

Script Turn on ECMAScript 2021 (ES12) mode

```
1 (function () {  
6 // Allow only EEE branch users to see EEE records  
7 if (gs.hasRole('bb1')){  
8     return true;  
9 }  
10 // Deny access for all others  
11 return false;  
12})();
```

Submit

Output Screenshots of Milestone 5

servicenow

All Favorites History Admin

Access Control - New Record

Search

Submit

Access Control

New record

Warning: A role, security attribute, data condition, script, or ACL control via reference needs to properly secure access with this ACL.

* Type

record

* Operation

delete

Decision Type

Allow If

Application

Global

Active

Advanced

Admin overrides

Protection policy

-- None --

* Name

Institution Details [u_institution_details]

Description

Applies To

No. of records matching the condition: 4

Add Filter Condition

Add OR Clause

-- choose field --

-- oper --

-- value --

Conditions

servicenow

All Favorites History Admin

Access Control - New Record

Search

Submit

Access Control

New record

Condition

All of these conditions must be met

-- choose field --

OR

AND

New Criteria

Data Condition

Condition

No. of records matching the condition: 4

Add Filter Condition

Add OR Clause

Branch

is

EEE

AND

OR

X

Advanced Condition

Controlled by References

Script

Turn on ECMAScript 2021 (ES12) mode

1

servicenow All Favorites History Admin : Access Control - New Record Search

Access Control New record

Advanced Condition

Controlled by References

Script ☒ Turn on ECMAScript 2021 (ES12) mode

```

1 (function () {
6   // Allow only EEE branch users to see EEE records
7   if (gs.hasRole('bb1')){
8     return true;
9   }
10  // Deny access for all others
11  return false;
12})();

```

Submit

Output Screenshots of Milestone 6

servicenow All Favorites History Admin : Access Control - New Record Search

Access Control New record

Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL

* Type record Application Global

* Operation delete Active ☒

Decision Type Allow If Advanced ☐

Admin overrides ☒

Protection policy -- None --

* Name Institution Details [u_institution_details] -- None --

Description

Applies To No. of records matching the condition: 4

Add Filter Condition Add OR Clause

servicenow All Favorites History Admin : Access Control - New Record Search

Access Control New record

Condition All of these conditions must be met

-- choose field --

OR AND

or

New Criteria

Data Condition

Condition No. of records matching the condition: 4

Add Filter Condition Add OR Clause

Branch is EEE AND OR X

Advanced Condition

Controlled by References

Script Turn on ECMAScript 2021 (ES12) mode

1

servicenow All Favorites History Admin : Access Control - New Record Search

Access Control New record

Branch is EEE AND OR X

Advanced Condition

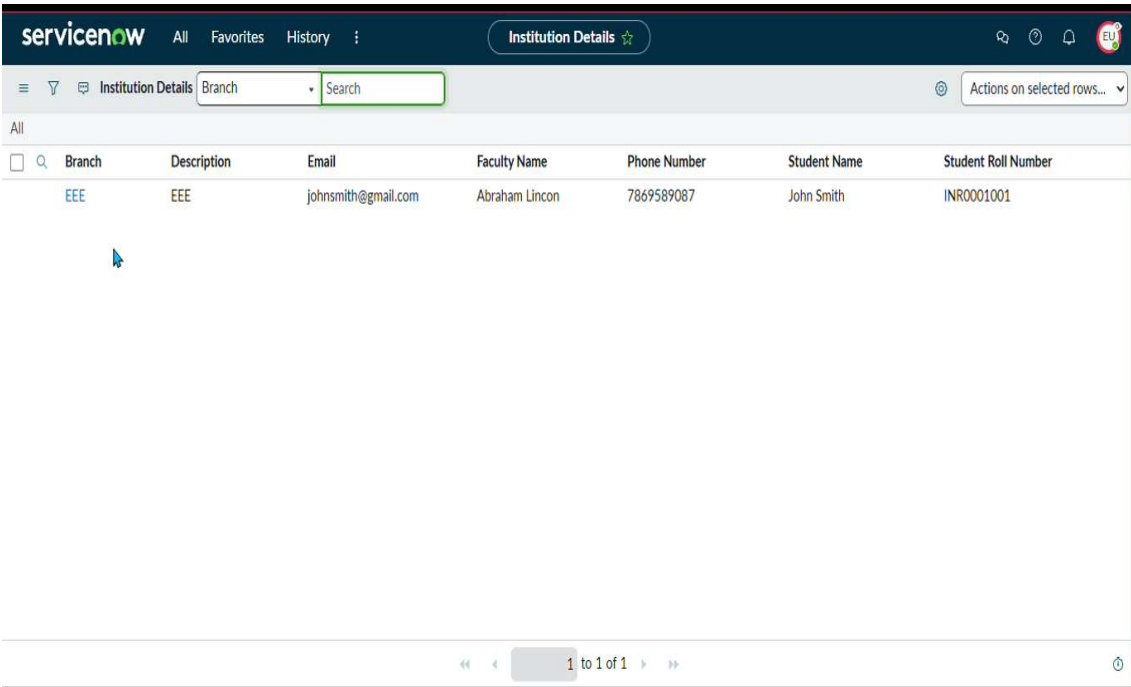
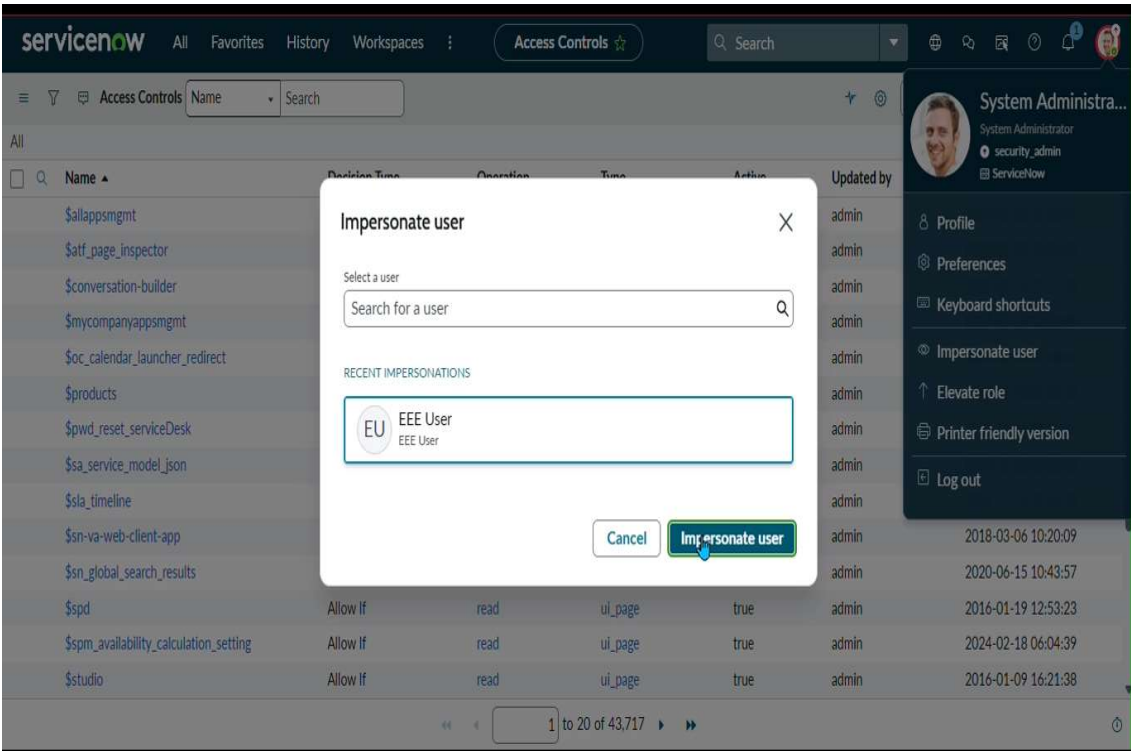
Controlled by References

Script Turn on ECMAScript 2021 (ES12) mode

```
1 (function () {
6   // Allow only EEE branch users to see EEE records
7   if (gs.hasRole('bb1')){
8     return true;
9   }
10  // Deny access for all others
11  return false;
12})();
```

Submit

Output Screenshots of Milestone 7



8. ADVANTAGES & DISADVANTAGES

Advantages

- ✓ Provides record-level security.
- ✓ Restricts unauthorized users from viewing records.
- ✓ Supports role-based access control.
- ✓ Uses Branch values as an additional access condition.
- ✓ Separately controls Read, Create, Write and Delete operations.
- ✓ Provides administrators with full access.
- ✓ Helps protect sensitive records from unauthorized modification or deletion.

Disadvantages

- ✓ ACL configuration requires careful role and condition management.
- ✓ Incorrect ACL configuration can result in access being allowed or denied incorrectly.
- ✓ Multiple ACLs must be maintained for different operations.
- ✓ Testing is required whenever roles, conditions or ACL scripts are changed.

9. CONCLUSION

The **Script-Controlled ACL – Restrict Record Access Based on Field Value** project demonstrates how ServiceNow ACLs can be used to enforce record-level security.

The project combines user roles and record field values to control access. EEE branch users can access EEE records, while administrators retain full access. Separate ACLs are used to control Read, Create, Write and Delete operations.

The project provides practical understanding of how ACLs protect records from unauthorized viewing, creation, modification and deletion.

10. FUTURE SCOPE

The project can be extended by introducing additional branch-based access rules, more granular role permissions and additional conditions for different types of records.

Future enhancements can also include more detailed access reports, centralized ACL monitoring and additional record-level security conditions.

11. APPENDIX

Source Code

Included the final ACL script and ACL configuration details used in the ServiceNow project.

```
(function () {  
  
    if (gs.hasRole('admin')) {  
  
        return true;  
  
    }  
  
    if (gs.hasRole('bb1')) {  
  
        return true;  
  
    }  
  
    return false;  
  
})();
```

Dataset Link

ServiceNow **u_institution_details** table.

Since this project uses **custom ServiceNow records** (ECE, EEE, CSE) rather than a separate public dataset, you can write:

GitHub & Project Demo Link

GitHub: Add project GitHub repository link here.

Project Demo: Add final project demonstration video/link here.

